Revision: 1.0 - 8 October 2026
Classification: Training Use Only

ASR-13 Primary Surveillance Radar

Representative dual-mode parameter datasheet for MATLAB agentic-workflow exercises

IMPORTANT - SIMULATION DATA ONLY

This fictional training artifact is designed for classroom radar modeling. It is not a manufacturer, FAA, airport, procurement, maintenance, navigation, or operational specification.

Configuration at a glance

Radar role	Primary surveillance
Coverage mode	Terminal / 360 deg
Baseline scan	12.5 rpm / 4.8 s revisit
Carrier frequency	2.80 GHz
Instrumented range	1.5-60 nmi
Target class	Fixed-wing aircraft

1. System identity and operating envelope

Parameter	Representative value
System designation	ASR-13 (training representative)
Radar type	Coherent pulsed primary surveillance radar
Frequency band	S-band
Instrumented range	1.5-60 nmi (2.8-111 km)
Azimuth coverage	0-360 deg mechanical scan
Elevation coverage	0-30 deg via six stacked 4.5 deg beams (exercise assumption)
Environmental baseline	Standard atmosphere; clear air; free-space baseline

2. Primary model input set - baseline surveillance mode

Parameter	Representative value
Carrier frequency, fc	2.80 GHz
Peak transmit power, Pt	1.10 MW
Pulse repetition frequency, PRF	1,000 Hz
Transmit pulse width, tau	0.80 us
Antenna peak gain, G	34 dBi
Azimuth 3 dB beamwidth	1.4 deg
Elevation 3 dB beamwidth	4.5 deg
Antenna scan rate	12.5 rpm (75 deg/s)
Receiver noise figure	4.0 dB
System losses	7.0 dB
Reference target RCS	10 m2

ASR-13 Modeling Notes

3. Detection and measurement assumptions

Parameter	Representative value	Use in exercise
Probability of detection, Pd	0.90 at reference condition	10 m2 target at 50 nmi, boresight, clear air.
Probability of false alarm, Pfa	1.0e-6 per resolution cell	Detector threshold evaluation constraint.
Range measurement accuracy	+/-90 m (1 sigma)	Add zero-mean Gaussian range error for reports.
Azimuth measurement accuracy	+/-0.12 deg (1 sigma)	Add zero-mean Gaussian bearing error for reports.
Range resolution (nominal)	120 m	Use c*tau/2; no pulse compression in baseline.
Unambiguous range (baseline)	150 km	At 1,000 Hz; exceeds the 60 nmi reporting gate.
Update interval	4.8 s	One measurement opportunity per antenna revolution.

4. Fast-update track mode - explicit advanced assumption

Parameter	Representative value	Modeling interpretation
Mechanical scan	60 rpm / 1 s revisit	Full 360 deg search refresh.
Physical azimuth beam	1.4 deg	About 3.9 ms dwell and four pulses at 1,000 Hz.
Scenario coverage sector	30 deg azimuth x 20 deg elevation	Operational coverage abstraction; not a 30 deg transmit beam.
Prioritized-track dwell	20 ms / 20 pulses	Electronically scheduled, phase-referenced dwell.
Search-mode integration	Four coherent pulses	Use for ordinary mechanical beam crossings.
Track-mode integration	20 coherent pulses	Use only when the scheduled dwell is active.

5. Data-quality and modeling rules

Keep physical beamwidth, coverage sector, revisit time, and coherent dwell as separate quantities. State target fluctuation, atmospheric, scan, CFAR, and detector-processing losses whenever a detection range is claimed. The baseline uses free-space propagation, boresight gain, a nonfluctuating 10 m2 target, and no CFAR loss unless a lesson explicitly adds them.